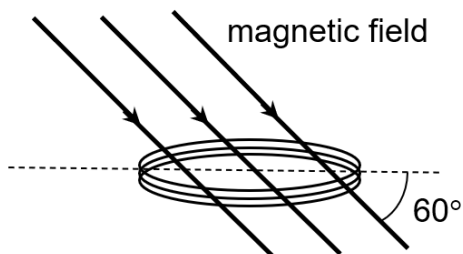


26

A magnetic field of uniform flux density 0.045 T is at an angle of 60° below the plane of horizontal coil of wire as shown.



The coil of wire has 4 turns and cross sectional area of 0.30 m^2 .

What is the magnitude of the change in magnetic flux linkage of the coil of wire when it is flipped over 180° about a horizontal axis?

A 0 Wb turns

B 0.012 Wb turns

C 0.035 Wb turns

D 0.094 Wb turns

Answer: D

Taking downwards as positive,

$$\begin{aligned}
 \Delta\phi &= \phi_f - \phi_i \\
 &= NA (B \sin 60 - (-B \sin 60)) \\
 &= 4 \times 0.3 \times [0.045 \sin 60 - (-0.045 \sin 60)] \\
 &= 0.094 \text{ Wb turns}
 \end{aligned}$$

27

The variation with time t of the voltage V of an alternating source applied across a 2.5Ω